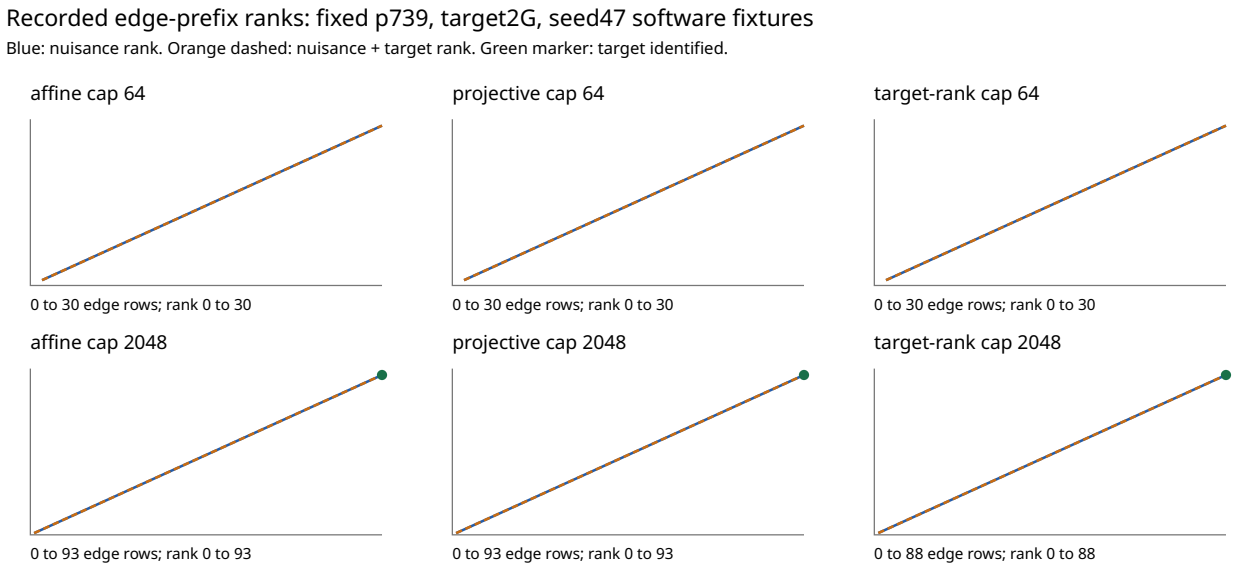
Signed collector stream replay and prefix information

Implemented a reusable offline Rust validator and CLI for signed collector JSONL. On six deterministic software streams, all 364 edge prefixes replayed successfully; six separate CLI invocations reproduced the complete prefix results. Twenty-one corruption and limit cases were rejected. The exact captured streams, expected fixture headers, per-prefix CSV, JSON outputs, source bindings and command receipts are included.



Fixture scope. One public synthetic instance: p=739, generator seed 23, target 2G, walk seed 47. The three variants are affine, projective and projective target-rank, each with candidate caps 64 and 2048. The smaller cap yields 30 edge prefixes in every variant; the larger yields 93, 93, 88. Larger-cap outputs recover scalar 2, checked on the source curve by the test. These are software controls, not the full 1,260-target cohort or performance measurements. Expected headers in these controls come from a separate native invocation; real callers must bind headers to independently pinned workload/custody.

Validated boundary. Reject duplicate JSON keys at every nesting level, unknown event fields, wrong expected header, nonconsecutive sequence, missing candidate disposition, invalid canonical residues, invalid node order, wrong first-duplicate references, graph-output/counter disagreement, incomplete terminal and post-terminal events. Nodes are checked as roots of the recorded divisor; edge root multiplicities, canonical point signs and destination curve equations are checked. A nonsplit disposition requires a nonsquare quadratic discriminant. Bounds cover input bytes, event/node counts, retained dense cells and cumulative diagnostic prefix cells, not total allocator overhead.

Information calculation. Replay preserves the signed equation su*xu+sv*xv=a+b*k. Node 0 is known zero and excluded from nuisance columns; new columns zero-extend older rows. The offline row is [A,-b,-a]. Each prefix computes rank(A), rank([A b]) and rank([A b c]); consistency and target identifiability agree with the native graph return in these controls. The diagnostic stays outside the producer because it changes shared operation counters.

Scope still open. Graph replay reuses the native SignedGraph; the rank diagnostic is a separate elimination implementation. Neither establishes an independently reviewed complete source-fidelity proof. Candidate walk labels, PRNG history and selection priority are not reconstructed; choice events receive reference/order checks only. Point/divisor checks do not prove the group-label invariant. Duplicate, identity and restart handlers exist but these six natural fixture streams do not exercise those event kinds positively. Independent review remains pending. No cohort cost ratio or observer-overhead estimate is produced.

Checks and failures. Final source-bound receipt: check-ivrc3mLQ. Two test functions passed, including six streams and 21 rejection cases; six CLI results match. Source identities before/after are unchanged. During editing, a patch-script assertion rejected a stale formatting match before writing; a subsequent check repeated unchanged code and is retained. All executed package checks passed. Prior root-library allocation failure and timeout remain unresolved; this package result does not replace full-repository validation.

Sources and continuation. Reused the published target-stream reconciliation source map, SignedGraph and the native observed collector, plus the existing rank-information diagnostic and its explicit sign convention. No new literature claim is made. Next technical work is to bind replayed candidates to the source walk-label invariant and check positive duplicate/identity/restart cases, while completing runtime admission for the unchanged 1,260-target campaign. New planning limits remain 25,200 worker-seconds ending 2026-11-03 06:00 UTC; this step launched no scientific worker. Canonical scientific result graphs remain unchanged.